

Question 1: (7 points) Given the following code Answer the following

```
class TypeA{
private:
    int x,*p; char c;
public:
    TypeA(int a)
    { x = a; }
    TypeA(int *q)
    { p = q; }
    TypeA(char c)
    { c = c; }
    void fun1(int a)
    { cout<<x+a; }
    int * fun2(int *k)
    {
        *k+=*p;
        return k; }
    char & fun3(char m)
    {
        char *g=new char(m);
        return *g; }
    int fun4(TypeA k)
    { return pow(*(k.p),2); }
    char & fun4(char &h)
    {
        if(h=='a')
            return ++h;
        else
            return --h; }
};
```

C.N.E
COMPUTER NETWORK
ENGINEERING

1- output if these instructions were executed in main TypeA xa(1),xb(new int(7)), xc('c'); int x=2; int *p=&x; p=xb.fun2(new int(2)); char cc='b'; cout<<xa.fun4(cc);	A) a
	B) b
	C) cc
	D) other
2- How many overloaded constructor in the class TypeA?	A) 4
	B) 5
	C) none
	D) other
3- which function returns a reference	A) fun1
	B) fun2
	C) fun3
	D) other
4- Which function returns a pointer?	A) fun1

	B) fun2
	C) fun3
	D) other
5- output if these instructions were executed in main TypeA xa(1),xb(new int(7)), xc('c'); cout<<xa.fun4(xb);	A) 16
	B) 49
	C) cc
	D) other
6- Output if these instructions were executed in main? TypeA xa(1),xb(new int(7)), xc('c'); int x=2; int *p=&x; p=xb.fun2(new int(2)); cout<<xc.fun3('d');	A) c
	B) d
	C) e
	D) other
7- what is the output if these instructions were executed in main TypeA xa(1),xb(new int(7)), xc('c'); int x=2; int *p=&x; p=xb.fun2(new int(2)); cout<<*p;	A) 9
	B) d
	C) 7
	D) other
Question 2: (4 points) What will happen in the following C++ code snippet?	
1- int a =100, b =200; int *p = &a, *q = &b ; p = q;	A) b is assigned to a
	B) p now points to b
	C) a is assigned to b
	D) q now points to a
2- int a = 5, b = 10, c = 15; int *arr[] = {&a, &b, &c}; cout << arr[1];	A) 5
	B) 10
	C) 15
	D) it will return some random number
3- char *ptr; char str[] = "abcdefg"; ptr = str; ptr += 5; cout << ptr;	A) fg
	B) cdef
	C) defg
	D) abcd
4- class test(int x,y;); main() {test t; t.x=1; t.y=t.x++; cout<<t.x<<t.y;}	A) 11
	B) 12
	C) error
	D) none
Question 3: (4 points) choose the correct answer	
1- A pointer is	A) the address of a variable
	B) an indication of the variable to be accessed next
	C) a variable for storing addresses
	D) the data type of an address variable
2- *test	A) refers to the contents of test
	B) dereferences test
	C) refers to the value of the variable pointed to by test

	D) all of the above
3- Definition for an array arr of 8 pointers that point to variables of type float is	A) *float arr[8];
	B) float* arr[8];
	C) float pointer[8];
	D) int *arr[8];
4- Under what conditions a destructor destroys an object?	A) Scope of existence has finished
	B) Object dynamically assigned and it is released using the operator delete
	C) Program terminated
	D) All of the above

Question 4: (5 points) Given the following namespace answer the following?

```
using namespace std;
namespace hhh{
    int * fun1(int *p){
        *p+=1;
        return p;}
    int & fun2(int &c){
        ++c;
        return c; }
    void fun3(int k){
        k+=2;
        cout<<k;
        return ; }
    namespace ddd{
        struct mystruct
        { string g="abc";
          int x=9;}xm; }
    class myc{ public: int x,y;};
}
```



C.N.E
COMPUTER NETWORK
ENGINEERING

1- int x=7; int*q=hhh::fun1(&x); cout<<x; cout<<*q;		2- int z=5; int &g=hhh::fun2(z); cout<<z; cout<<g;	
3- int k=5; hhh::fun3(k); cout<<k;		4- hhh::ddd::mystruct xk; cout<<xk.g; cout<<hhh::ddd::xm.x;	
		5- using namespace hhh; myc xc; xc.x=2; xc.y=1; cout<<xc.x+xc.y+ hhh::ddd::xm.x;	
Question 1:	1- A	2- D	3- C
Question 2:	1- B	2- D	3- A
Question 3:	1- D	2- B	3- D
Question 4:	1- 88	2- 66	3- 75

4- B 5- B 6- B 7- A
4- C 4- D
5- abc9 12